Business Management HL · Set A · Paper 2

Worked mark scheme · 105 minutes · 50 marks

Award equivalent valid methods and conclusions. Apply the indicated mark allocation to each part. Credit consistent follow-through where the subsequent reasoning remains valid.

Section A · Polder Panels [12]

(a) **[4]**

Calculate the current ratio and the acid-test ratio.

1 mark · Divide 540,000 by 300,000.

1 mark · Current ratio is 1.80:1.

1 mark · Subtract inventory before dividing by current liabilities.

1 mark · Acid-test ratio is 1.10:1.

(b) **[4]**

Calculate debtor days. Estimate how much cash would be released if debtor days fell to 40, assuming credit sales were unchanged.

1 mark · Debtor days equal 220000/1460000 times 365.

1 mark · 55 days.

1 mark · Average daily credit sales are 4,000, so target receivables are 160,000.

1 mark · CU 60,000 would be released.

(c) **[4]**

Explain two possible consequences of tightening customer credit for Polder.

1 mark · Faster receipts support machine purchases or supplier payments.

1 mark · This reduces reliance on an overdraft.

1 mark · Its largest customer may place orders elsewhere.

1 mark · Lower sales could reduce contribution and offset the liquidity gain.

Section A · Tern Works [12]

(a) **[3]**

Calculate the payback period in years.

1 mark · CU 30,000 remains after two years.

1 mark · A further 30,000/70,000 of a year is needed.

1 mark · Payback is approximately 2.43 years.

(b) **[4]**

Calculate the net present value using the supplied factors.

1 mark · Discount the three inflows separately.

1 mark · Present values are CU 49,995, 53,690 and 52,570.

1 mark · Subtract the initial outlay from their sum.

1 mark · NPV is CU 6,255.

(c) **[3]**

Find the lowest year-3 net cash inflow that gives a non-negative NPV, holding years 1 and 2 unchanged. Explain the significance for Tern.

1 mark · Set $49995 + 53690 + 0.751C = 150000$.

1 mark · $C = 46315 / 0.751 \approx 61671.1$, so at least CU 61,672 if using whole units.

1 mark · A shortfall of about CU 8,329 against the forecast removes the positive NPV; losing the contract could make the investment unacceptable.

(d) **[2]**

Explain one difference between the information supplied by NPV and by payback.

1 mark · Payback measures how quickly the initial outlay is recovered.

1 mark · NPV values all forecast cash flows at the discount rate, including amounts received after payback.

Section A · Fenn Freight [6]

(a) **[3]**

Using expected monetary values, determine which option is preferred.

1 mark · Lease EMV is $0.55(140000) - 0.45(50000)$.

1 mark · CU 54,500.

1 mark · Pilot EMV is $0.55(65000) + 0.45(15000) = 42500$, so the lease leads by CU 12,000.

(b) **[3]**

Find the high-demand probability at which the two options have the same EMV. Explain one reason a manager might still select the pilot at probability 0.55.

1 mark · Set $190000p - 50000 = 50000p + 15000$.

1 mark · $p = 13/28 \approx 0.4643$.

1 mark · A CU 50,000 loss could threaten liquidity, whereas the pilot offers a positive outcome in both scenarios.

Section B · Option 1 · Koru Packaging [20]

(a) **[2]**

Calculate current capacity utilization.

1 mark · Divide output by rated capacity.

1 mark · 80%.

(b) **[4]**

Calculate expected saleable output before and after the change and the expected monthly gain in contribution after maintenance costs.

1 mark · Currently 121,600 cartons.

1 mark · Forecast 132,300 cartons.

1 mark · Increment is 10,700 × 1.20 minus 8,000.

1 mark · CU 4,840 per month.

(c) **[4]**

Find the highest reject rate that would meet the customer's monthly target at the forecast output. Explain why this threshold matters.

1 mark · Set $135000(1-r) \geq 130000$.

1 mark · $r \leq 1/27 \approx 3.70\%$.

1 mark · A reject rate above this leaves too few saleable cartons.

1 mark · Two months of shortfalls could lose the contract despite a profitable average month.

(d) **[10]**

Evaluate adopting the preventive maintenance programme.

10 marks · Current output misses the target by 8,400 cartons. Expected saleable output after maintenance exceeds the target by only 2,300. The expected gain is CU 4,840, but reliability and the spread of reject rates matter under the cancellation clause. Trial the programme, monitor rejects and establish backup stock or delivery tolerances. Full adoption can be justified if forecast output is repeatable and inventory costs are affordable. Use best fit across the response. 0 gives no creditable response. At 1–2, identify a relevant issue with little development. At 3–4, explain a point with limited case evidence. At 5–6, develop a relevant argument using case evidence. At 7–8, weigh competing arguments using specific evidence and reach a supported judgement. At 9–10, sustain a balanced case-specific judgement, explain a material limitation or condition and show how it affects the decision. Credit a different conclusion when the analysis supports it.

Section B · Option 2 · Reed Rail Kitchens [20]

(a) **[2]**

Calculate the budgeted overhead absorption rate per labour hour.

1 mark · Budgeted hours are $18000(0.4) = 7200$.

1 mark · CU 30 per labour hour.

(b) **[4]**

Calculate full cost per kit at the budgeted rate and incremental profit from the extra order.

1 mark · Unit overhead is 0.4 × 30 = CU 12.

1 mark · Full cost is CU 26.

1 mark · Incremental profit is $3000(28-14) - 6000$.

1 mark · CU 36,000.

(c) **[4]**

Determine the maximum number of regular retail kits that could be displaced before the order ceases to increase profit. Explain the relevance of spare capacity.

1 mark · Lost contribution is CU 20 per displaced kit.

1 mark · 1,800 displaced kits give zero incremental profit, so at most 1,799 for a strictly positive gain.

1 mark · The order takes 1,200 hours, within the 1,600 spare hours overall.

1 mark · Peak-time constraints can still cause displacement despite annual spare capacity.

(d) **[10]**

Discuss whether Reed should accept the rail contract on the terms offered.

10 marks · The CU 36,000 incremental contribution and use of spare capacity favour acceptance. Allocated overhead is already incurred. Peak priority could displace retail customers, damage availability and reduce contribution by CU 20 per lost sale. Negotiate delivery windows or an opportunity-cost surcharge and test peak rosters before accepting. A rejection is supported if the delivery promise threatens established customers. Use best fit across the response. 0 gives no creditable response. At 1–2, identify a relevant issue with little development. At 3–4, explain a point with limited case evidence. At 5–6, develop a relevant argument using case evidence. At 7–8, weigh competing arguments using specific evidence and reach a supported judgement. At 9–10, sustain a balanced case-specific judgement, explain a material limitation or condition and show how it affects the decision. Credit a different conclusion when the analysis supports it.

End of mark scheme · 50 marks